

EXPLAINABLE AI FOR EDUCATORS

Engaging Students and Inspiring Entrepreneurial Mindsets

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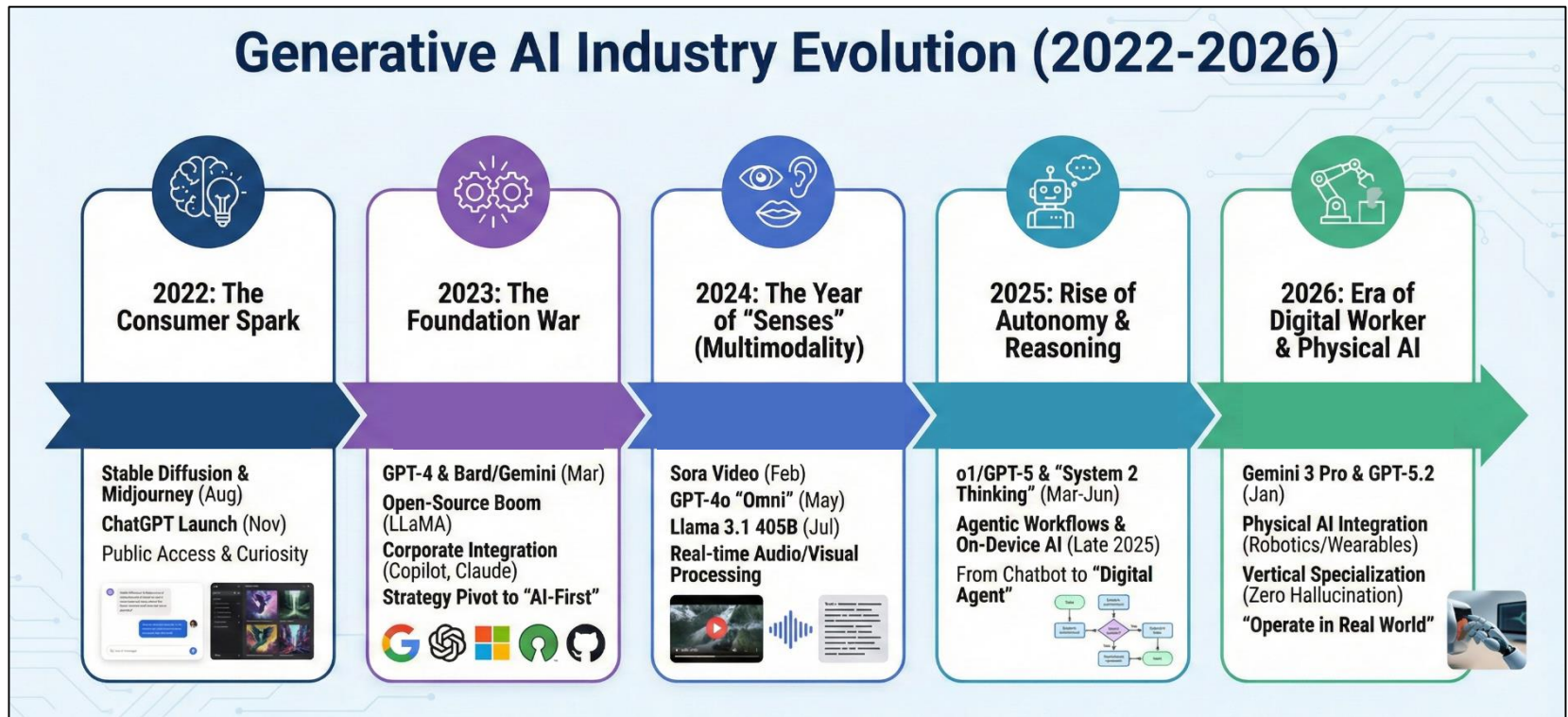
Biomedical Engineering

Participant Survey

Background experience/knowledge?



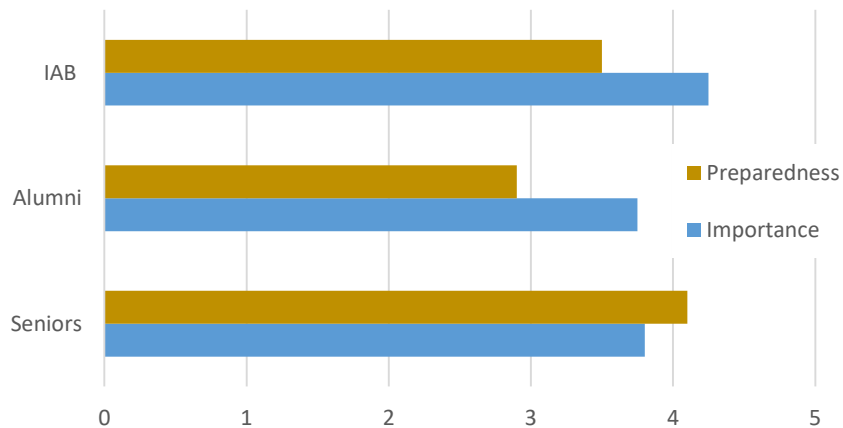
GenAI Background



- Approximately 1 in 6 people worldwide use GenAI tools.
- 74% of US adults under 30 use GenAI.
- 79% of businesses have integrated GenAI into at least one workflow.

AI Fluency

BME Essential Skills Survey: AI Fluency



- AI does not benefit all students **equally**
- Generic workshops alone are **insufficient**
- Faculty development is **important**

ANTHROPICAnthropic AcademyCourses

**AI Fluency: Framework & Foundations**
Learn to collaborate with AI systems effectively, efficiently, ethically, and safely

**AI Fluency for educators**
This course empowers faculty, instructional designers, and educational leaders to apply AI Fluency into their own teaching practice and institutional strategy.

**AI Fluency for students**
This course empowers students to develop AI Fluency skills that enhance learning, career planning, and academic success through responsible AI collaboration.
<https://www.anthropic.com/>

Assessments with GenAI

Students can generate essays or code in seconds, so traditional take-home assignments are no longer reliable indicators of learning.

- Verify understanding with oral assessments, in-class writing and step-by-step submissions.
- Ask students to reference class materials and discussions, or unique datasets that AI tools don't normally access.
- Update assessment rubrics to prioritize analytical depth, originality, transparency of process and intellectual engagement.

[Teaching in the age of generative AI: why strategy matters more than tools.](#)

Goal of the Workshop

- Using Explainable AI (XAI) in EML class modules
- Demonstration of existing tools in technical and engineering design topics
- Create XAI learning objectives for your courses



Agenda at a Glance



Intro to EML



**Chain of Thoughts
Activity**



SHAP Tool Activity



Co Creation with AI



Intro to EML

ENTREPRENEURIAL MINDSET + ENGINEERING SKILLSET = EDUCATIONAL OUTCOMES

ENTREPRENEURIAL MINDSET

CURIOSITY

DEMONSTRATE constant curiosity about our changing world

EXPLORE a contrarian view of accepted solutions

CONNECTIONS

INTEGRATE information from many sources to gain insight

ASSESS and **MANAGE** risk

CREATING VALUE

IDENTIFY unexpected opportunities to create extraordinary value

PERSIST through and learn from failure

COUPLED WITH

ENGINEERING THOUGHT AND ACTION

APPLY creative thinking to ambiguous problems

APPLY systems thinking to complex problems

EVALUATE technical feasibility and economic drivers

EXAMINE societal and individual needs

EXPRESSED THROUGH

COLLABORATION

FORM and **WORK** in teams

UNDERSTAND the motivations and perspectives of others

AND

COMMUNICATION

CONVEY engineering solutions in economic terms

SUBSTANTIATE claims with data and facts

AND FOUNDED ON

CHARACTER

IDENTIFY personal passions and a plan for professional development

FULFILL commitments in a timely manner

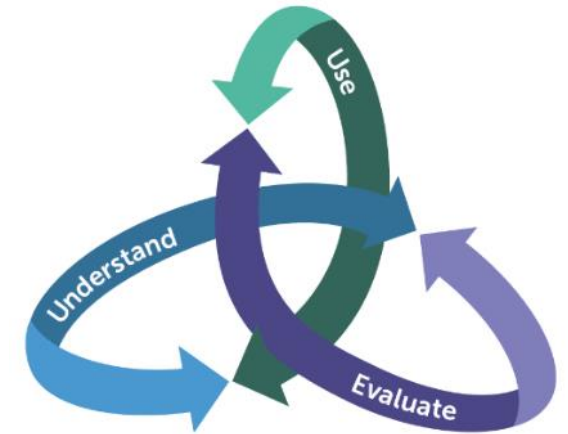
DISCERN and **PURSUE** ethical practices

CONTRIBUTE to society as an active citizen

Engineering Education

Dual mandate

1. Acquire deep domain knowledge
2. Develop critical thinking



AI Literacy Framework



- Students risk cognitive offloading, where they passively accept AI outputs
- "Human-in-the-Loop" pedagogical model

Entrepreneurial Minded Learning

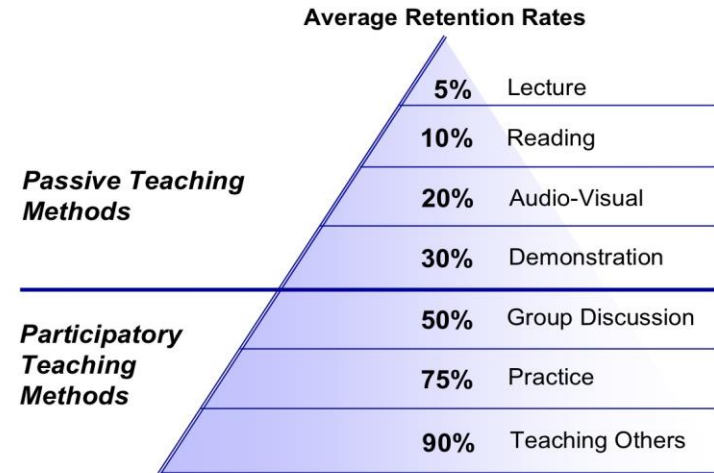
**ENGINEERING
UNLEASHED** 
POWERED BY KEEN



MINDSET + SKILLSET
A TANDEM DEVELOPMENT

<https://engineeringunleashed.com>

The Learning Pyramid*



*Adapted from National Training Laboratories. Bethel, Maine



Curiosity

Understand the broader world. Look forward. Explore multiple perspectives.



Connections

Think outside the box. Place old ideas in new contexts. Gain insights.



Creating Value

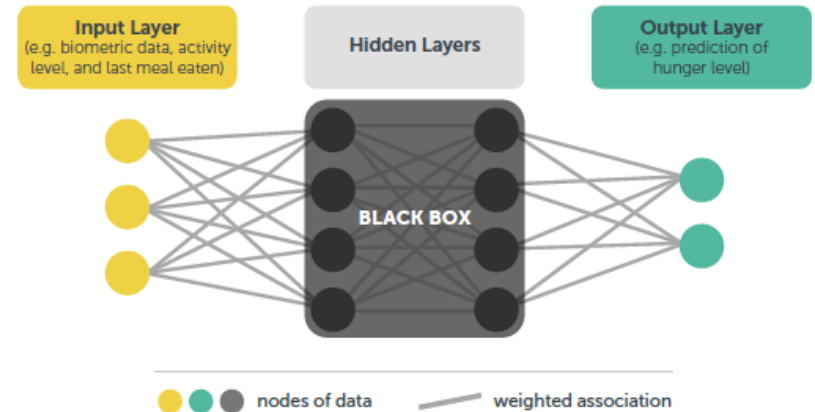
Seek opportunities. Understand stakeholders. Have a positive impact.

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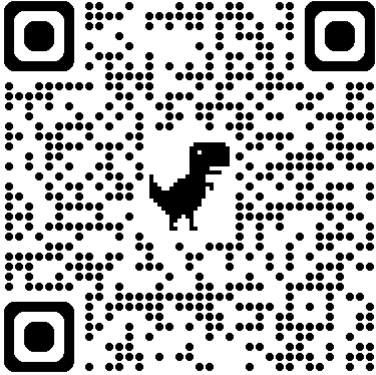
What is XAI?

Make the "thought process" of machine learning models transparent/understandable to humans



- Interpretability and Explainability
- Global or Local
- Ante-hoc vs. Post-hoc

What is XAI?



Google Scholar | Labs

what is explainable artificial intelligence

Current session

what is explainable artificial intelligence

✓ Found 10 relevant results

+ New session

Scholar Labs is designed for detailed research questions. Some questions to explore:

→ What are the diverse conceptual frameworks and definitions underpinning explainable artificial intelligence?

→ How do different XAI methodologies impact user trust and system adoption across various domains?

→ What are the current technological limitations and challenges facing the practical implementation of XAI?

[HTML] Explainable Artificial Intelligence (XAI): Concepts, taxonomies, opportunities and challenges toward responsible AI

[HTML] sciencedirect.c... Find It @ Lawrence T...

AB Arrieta, N Díaz-Rodríguez, J Del Ser, A Bannetot... - Information fusion, 2020 - Elsevier

Examines existing literature and contributions related to eXplainable AI (XAI), which addresses the inherent problem of explainability in the latest sub-symbolic Machine Learning techniques like ensembles or Deep Neural Networks.

- **Defines Explainable ML:** Establishes a novel definition of explainable Machine Learning, emphasizing the audience for which the explanation is sought, and provides a thorough taxonomy to stimulate future research.
- **Presents Model Taxonomies:** Discusses a taxonomy of recent contributions related to the explainability of various Machine Learning models, including a dedicated taxonomy for explaining Deep Learning methods.

☆ Save 99 Cite Cited by 10829 Related articles All 21 versions

[PDF] Four principles of explainable artificial intelligence

[PDF] nist.gov

PJ Phillips, PJ Phillips, CA Hahn, PC Fontana... - 2021 - nvlpubs.nist.gov

Introduces four principles—explanation, meaningful, explanation accuracy, and knowledge limits—that constitute the fundamental properties of Explainable AI (XAI) systems.

- **XAI System Requirements:** Proposes that XAI systems must provide evidence or reasons for outcomes, explanations that are understandable to users, and explanations that accurately reflect the system's process.
- **XAI Theories and Concepts:** Presents five categories of explanation and summarizes the multidisciplinary theories and core concepts underlying explainable AI, including computer science, engineering, and psychology.

Ask a follow-up

Scholar Labs is experimental and can make mistakes. [Help & Privacy](#)

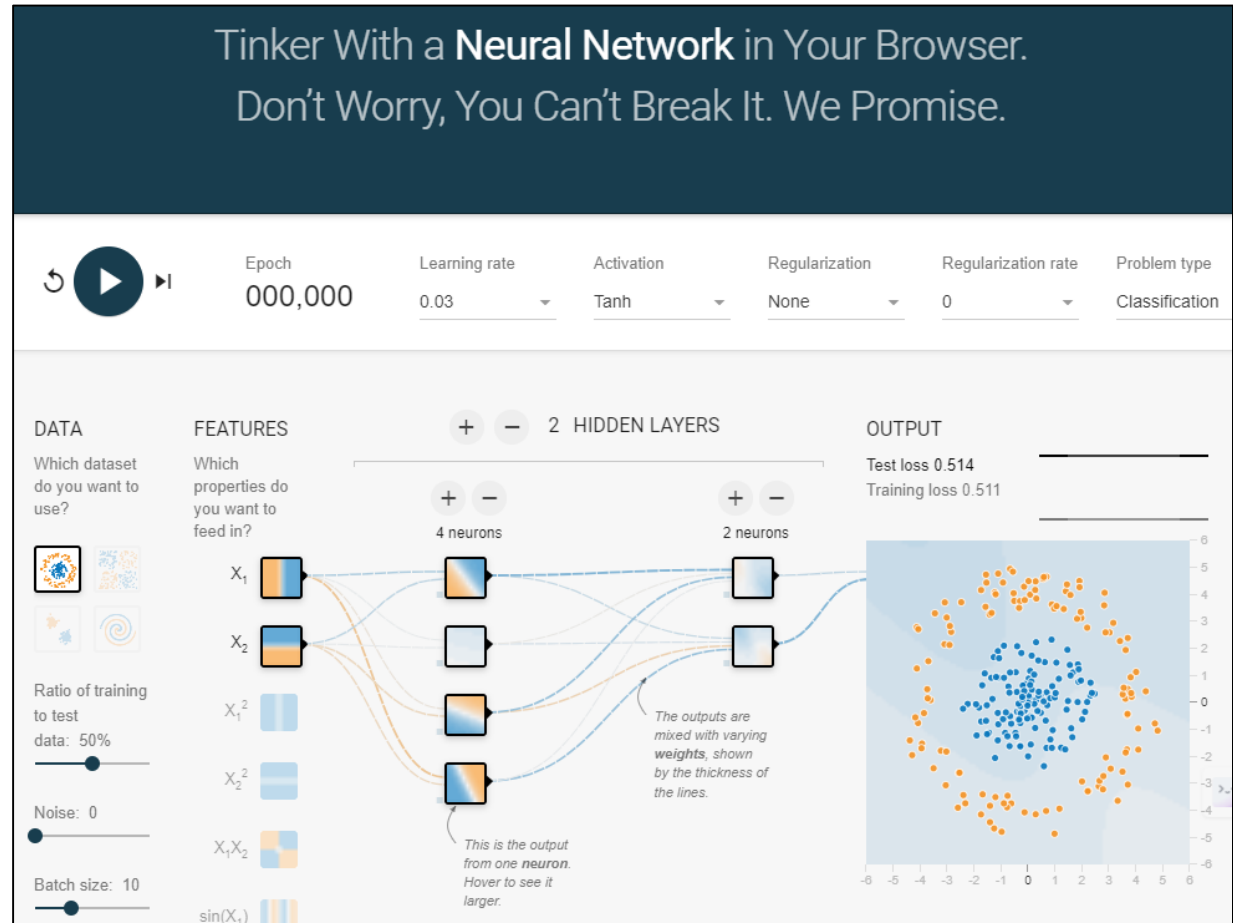
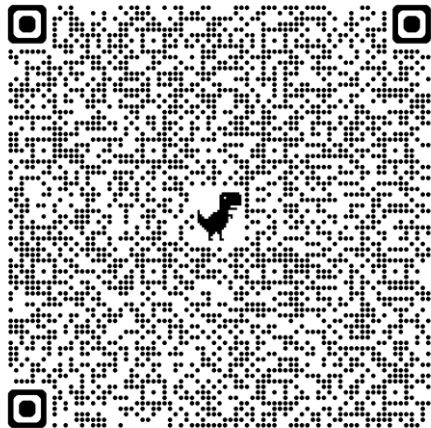
Safety Critical Decisions



NVIDIA DRIVE AV ([Alpamayo](#)) uses an AI end-to-end stack for core driving, alongside a parallel classical safety stack that adds redundancy and safety guardrails.

[NVIDIA at CES 2026: January 5 | Las Vegas, NV](#)

Hands on with Machine Learning



<https://playground.tensorflow.org/>

Generative AI Basics

The Basics

- Just Ask

Can it do [X]? Better Prompt? Sources?

Follow Ups

Is that true?

Am I missing something?

What Could Go Wrong?

Grade my work (make an outline/rubric)

If You Ask For Something Imaginary, It Will
Imagine It

Generative AI Basics

The Basics

- Just Ask
- It Can't Read Your Mind

Engineer Context, not the Prompt

Avoid zero-shot, use examples

Narrow your task, avoid context overload

It can Read Between The Lines (being explicit is better)

Custom Instructions / Memory / Privacy Settings

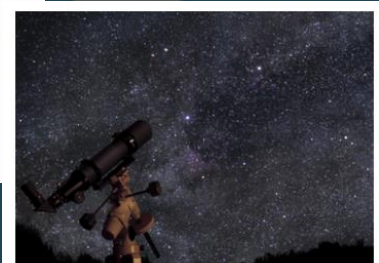
Generative AI Basics

The Basics

- Just Ask
- It Can't Read Your Mind
- Use The Right Tool

- Search
- Deep Research
- Canvas/Workspace
- Image/Video
- Agents
- Scheduled Tasks
- Connectors/Model Context Protocol
- Study and Learn/Guided Learning
- Projects/Notebooks/Spaces

John Peponis, Assistant Professor of Practice
LTU Biomedical Engineering



Curiosity

Understand the broader world. Look forward. Explore multiple perspectives.

AI-Enabled Toolkits

I am creating a lecture for engineering students in a design class and for my slides. I want to include tools that are general to the sum of add to my

Revolutionary AI-Enabled Toolkits



AI Startup Toolkit (2026 Edition)



I. Design & Generative Engineering

- **Neural Concept:** Deep learning for near-instant physics (CFD/FEA) simulations.
- **Autodesk Forma:** AI-driven conceptual design for site & environmental analysis.
- **Spline AI:** Real-time 3D modeling & spatial design via natural language.



II. Rapid Software & Hardware Prototyping

- **Cursor / Windsurf:** "Agentic" IDEs that understand entire codebases & build features.
- **Figma AI:** Instant UI/UX layouts that convert directly into production-ready code.
- **KiCad + AI Plugins:** Open-source PCB design with AI-assisted routing & components.



III. Research & Market Validation

- **Perplexity Pro:** Real-time technical research & patent-landscape synthesis.
- **Glean:** AI search for internal knowledge, perfect for managing engineering specs.
- **Flowcase:** Automates technical proposals & RFP responses for contracts.



IV. Brand & Growth (The "Pitch")

- **Design.com:** All-in-one AI suite for branding, logos, & web presence.
- **Runway ML / Sora:** Text-to-video for high-fidelity product explainers & demos.
- **ElevenLabs:** Human-quality voiceovers for global marketing in 30+ languages.

Gemini. (2026, January 21). *General Themes of AI Toolkits.*

(Startuptools.ai, Zapier)

Why It Matters.

Tout bon travail attendants plus a
Création de votre plan de marketing

Gemini. (2025, August 26). *General Themes of AI Toolkits.* [Large language model].

<https://gemini.google.com/>

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BME 5313

Engineering Applications in Orthopedics

- Orthopedics Industry
- Simulation Active Learning
- Weekly Journal Club
- Peer-reviewed Research Paper
- Opportunity Creation Project
- Medical Devices Regulation
- Clinical/Hospital Tours

BME 5305 Orthopedics Take Home Final

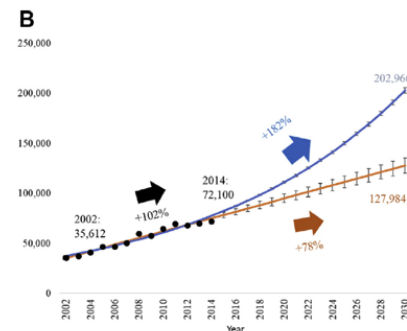
Due: Thursday 5/8/2025 3:45 pm

For the following questions, use generative AI (Perplexity, NotebookLM, or others) to suggest topics of interest and comparison reference articles. Submit the link to your AI discussion as well as **your own essay responses** (paragraph formatting) that includes adequate explanation, background information, referenced sources, and reflection on the impact of Biomedical Engineering for each question. The total length of your responses to the questions should be 2-3 pages each.

Generative AI Discussion Links (25 points) - be sure to follow-up on specific details, reference information or ask for review of other information sources (uploaded documents).

1. (25 pts) The chart below (from 2020) shows projected ranges for the number of revision total knee arthroplasty procedures expected through 2030 from <https://www.sciencedirect.com/science/article/pii/S0883540320301911>

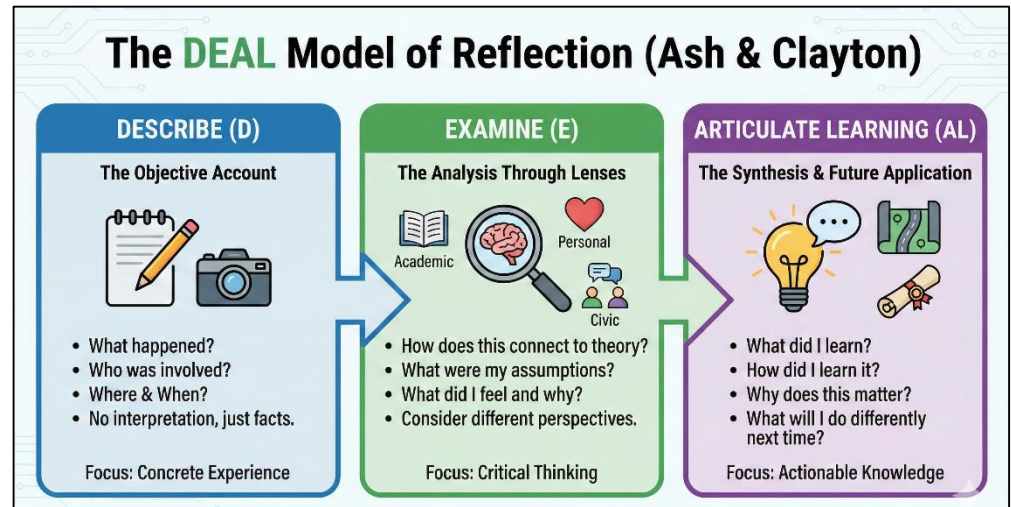
Modern technology, such as ultra-high molecular weight polyethylene, newer tibial polyethylene geometries, component design to improve prosthetic stability, additives such as vitamin E and advancements in bearing surfaces may increase success rates, however even modern implants and well-fixed components have a finite life span that is more likely to be exceeded in the younger active patient.



Based on the topics discussed this semester and your knowledge from Biomedical Engineering, **describe one design constraint** for total knee arthroplasty and **how this impacts the number of revision surgeries**.

Reflection Based Assignments

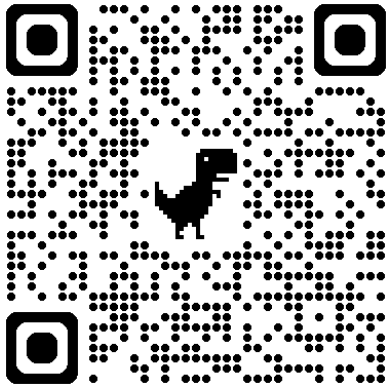
Student Centered
Collaborative
Structured Yet Flexible



Gemini. (2026, January 14). *Graphic for the DEAL model.*

- **Reflection**: What, So-what, Now-what?
- **Connect** topics to prior knowledge and spark curiosity.
- **Metacognition**: How do I know this?

Critical Reading Protocol



Orthopedics Journal Club Critical Reading Protocol

Create notebook

Analyt

Sources

+ Add sources

Try Deep Research for an in-depth report and new sources!

Search the web for new sources

Web Fast Research

Select all sources

Costa2022_EffectsInitialGraftTension12...

Dr Plopper's How to Read a Paper.pdf

ImprovingJournalClubPresentations.pdf

JournalClubAssignment.docx

Chat

Guide me through the Costa 2022 article based on the Dr. Plopper reading method so that I can create an effective journal club discussion for my Engineering Applications in Orthopedics course.

To guide you through the Costa 2022 article for your Engineering Applications in Orthopedics journal club, I have structured this analysis using **Dr. Plopper's Guide to Critical Reading of Primary Literature**. This method emphasizes deconstructing the paper in a specific order—starting with the title and ending with a future projection—rather than reading linearly from start to finish 1.

Here is your step-by-step guide to leading the discussion.

Step 1: Deconstruct the Title

Goal: Understand every word and predict the experiments before reading the text 2 3.

- The Title:** "Effects of Initial Graft Tension and Patient Sex on Knee Osteoarthritis Outcomes After ACL Reconstruction: A Randomized Controlled Clinical Trial With 10- to 12-Year Follow-up" 4.
- Engineering Context:** The key engineering variable here is "**Initial Graft Tension**." As engineers, your group should immediately ask: How is tension defined mechanically (Newtons vs. strain)? How is it applied?
- Predictions to discuss:** Based on the title, you should expect the paper to compare two distinct mechanical states (high vs. low tension) and their long-term effect on joint degeneration (osteoarthritis) 5. You should also expect a biological comparison between male and female subjects 5.
- Plopper's Tip:** Write down a list of expected experiments or data visualizations (e.g., "I expect a graph showing cartilage wear vs. tension force") before scrolling down 3 6.

Step 2: Read the Abstract

Start typing... 4 sources

NotebookLM can be inaccurate; please double check its responses.

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<https://notebooklm.google.com/notebook/29cec7e7-5246-4a51-93c9-5bf396097a3f>

Improving JC Discussions

What are three engaging discussion questions from this article that fit the template from the assignment requirements?

Based on the **Costa 2022** article and the specific templates provided in your **Journal Club Assignment** requirements **1**, here are three engaging discussion questions designed to stimulate critical thinking regarding engineering applications and experimental design.

1. Applications in Surgical Engineering

Question: "What are the applications of the results presented in this paper to the field of **surgical instrumentation**, specifically regarding the authors' use of 'laxity-based' tensioning versus quantitative force measurements?"

• Context for the Discussion Leader:

- **The Issue:** The authors used a "laxity-based" approach, tensioning the graft to match the uninjured knee's laxity rather than applying a specific numeric force (e.g., 80 Newtons) **2**.
- **Engineering Discussion:** Ask your classmates if "matching the feel" of the other knee is a reliable engineering standard. Since the study found no significant difference between tension groups **3**, does this suggest that expensive, high-precision tensioning devices are unnecessary? Or, conversely, does the high rate of osteoarthritis in *both* groups **4** suggest that we need *better* sensors to find an optimal tension that "laxity matching" fails to achieve?

2. Experimental Design and Correlation

Question: "What other experiments could be used to bolster the authors' claims that **higher physical activity levels** cause the **increased radiographic osteoarthritis** observed in male patients?"

• Context for the Discussion Leader:

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NotebookLM Studio

Journal Club Flash Cards

Based on 3 sources

In the Costa et al. study, how was the 'low-tension' ACL graft fixation defined?

The graft was tensioned with the knee at 0° of flexion to match the antero-posterior (AP) laxity of the contralateral knee.



Studio > Audio Overview

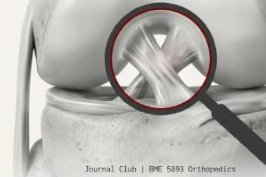
Activity Not Tension Drives ACL Arthritis

2 hosts



The Case of the Aching Knee: A 12-Year Investigation into Osteoarthritis After ACL Surgery

A Critical Appraisal of Costa et al., AJSM 2022



Journal Club | BPE 5893 Orthopedics

AI NotebookLM

Case File #AC1001000007



Study Type: A Level 1 Evidence Randomized Controlled Trial (RCT). The gold standard for a therapeutic question.

Publication: The American Journal of Sports Medicine, 2022.

Authors: Costa, M.Q., Fleming, B.C., et al. from Brown University / Rhode Island Hospital.

Relevance: This is the long-term (10-12 year) follow-up of a previously published cohort, allowing us to see the true endgame of the surgical variables. PTOA is a progressive disease, so long-term data is critical.

AI NotebookLM

The Unsolved Mystery of Post-Traumatic Osteoarthritis (PTOA)



The Problem: Over 400,000 ACL injuries occur annually in the U.S.

The Intervention: ACL Reconstruction (ACLR) is the standard of care to restore knee function.

The Paradox: Despite successful surgery, a high percentage of patients develop debilitating osteoarthritis 10 to 15 years later. ACLR fails to prevent PTOA.

The Question: Why does this happen? What factors in the operating room could be influencing the joint's fate a decade later?

AI NotebookLM

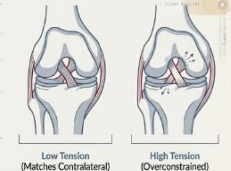
The Prime Suspect: Initial Graft Tension

Population: In patients undergoing ACLR...

Intervention: ...does a low initial graft tension...

Comparison: ...compared to a high initial graft tension...

Outcome: ...affect the 10-12 year outcomes of Post-Traumatic Osteoarthritis, knee function, and patient-reported scores?



- **Secondary Question:** Does patient sex play a role?
- **The Stated Hypothesis:** The investigators predicted no differences based on graft tension or sex.

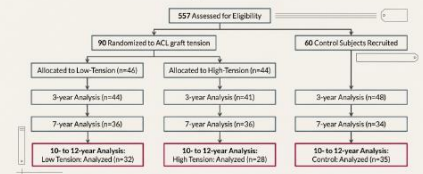
AI NotebookLM

The Investigation Protocol (PICO)

P: Patients	I: Low Tension	C: High Tension	O: Outcomes (10-12 Years)
90 patients randomized (final analysis n=60). Unilateral ACL injury, age 15-50. Matched with 60 uninjured controls.	Graft tensioned at 0° flexion to match the laxity of the healthy contralateral knee.	Graft tensioned at 30° flexion to be 2mm tighter (overconstrained) than the healthy knee.	A multi-faceted evidence-gathering approach: - Patient-Reported: KOOS (Knee Injury and Osteoarthritis Outcome Score) - Clinical Exam: IKDC Score - Imaging (The Hard Evidence): OARSI (X-ray score) & WOMPS (MRI score for whole-organ assessment).

AI NotebookLM

Profiling the Subjects



****Key Takeaway:** Loss to follow-up is a major challenge in long-term orthopedic trials, but no significant difference in dropout rates was seen between groups.

AI NotebookLM

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How could **GenAI** tools be used to increase students' reflection?

Choose a technical learning outcome (LO) from your course



Pick an assignment related to this LO from your course/project

Which EM learning outcomes best align with your revised course assignment?

Curiosity

- a. Develop a propensity to ask MORE questions.
- b. Be able to formulate SALIENT questions.
- c. Question information that is given without sufficient justification.
- d. Collects feedback and data from many customers and customer segments.
- e. Recognize and explore knowledge gaps.
- f. Critically observes surroundings to recognize opportunity.
- g. View problems with an open mindset and explore opportunities with passion.
- h. Be able to self-reflect and evaluate preconceived ideas, thoughts, and accepted solutions.
- i. Explores multiple solution paths.
- j. Gathers data to support and refute ideas.
- k. Suspends initial judgement on new ideas.
- l. Take ownership of, and express interest in topic/expertise/project.
- m. Observes trends about the changing world with a future-focused perspective.



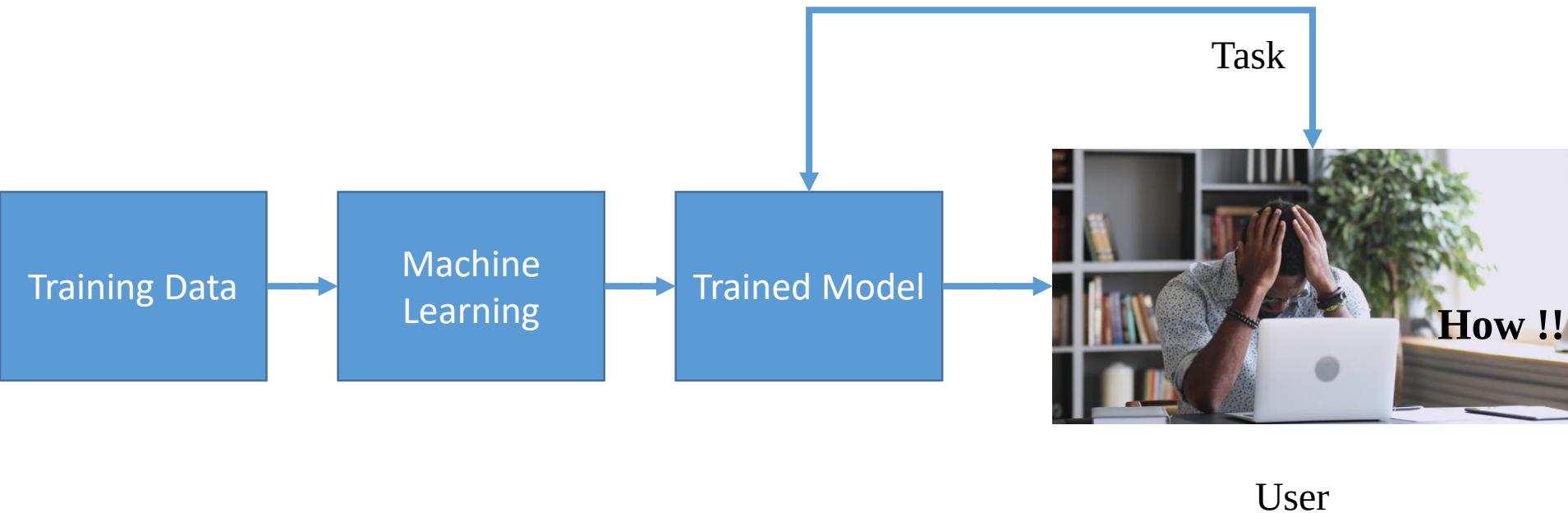
SHAP Tool Activity



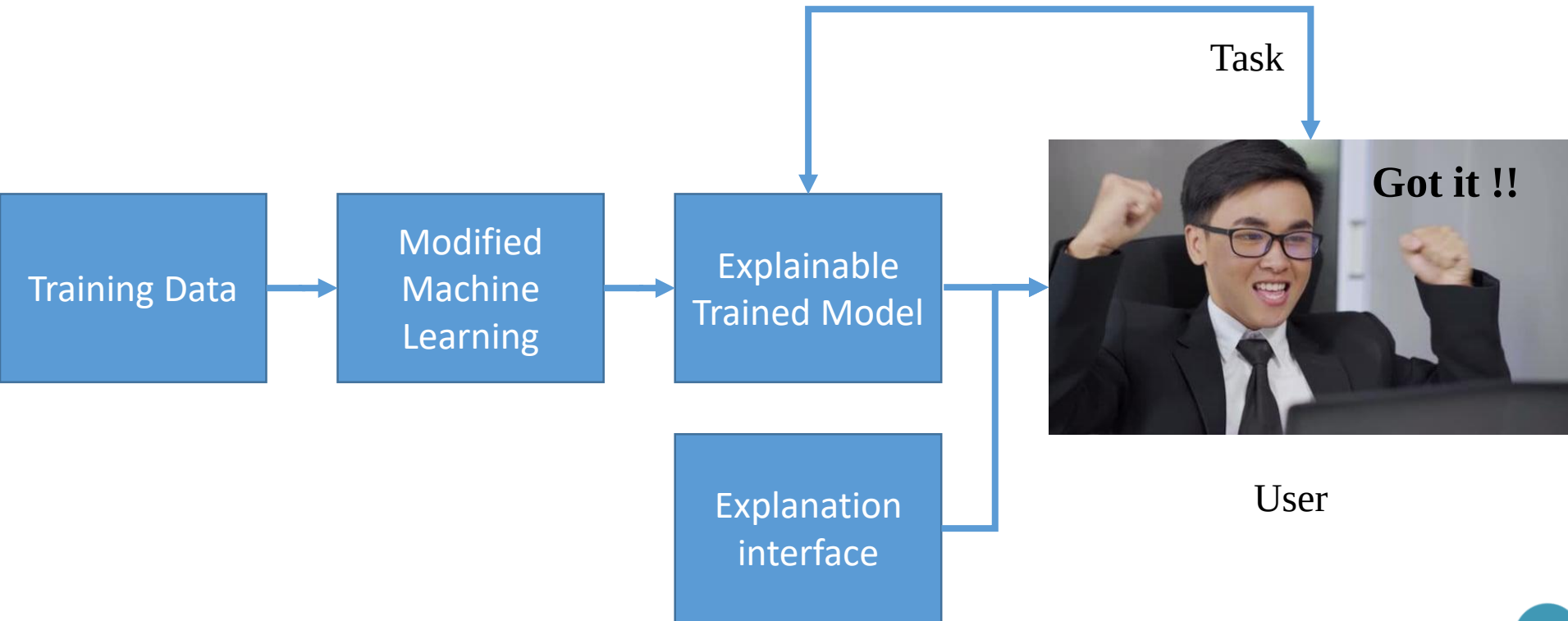
Connections

Think outside the box. Place old ideas in new contexts. Gain insights.

Classical Machine Learning Architecture



Machine Learning Architecture with xAI



Group Activity

Start by Introducing Yourself to your table...

- Institution
- Role
- Familiarity with KEEN from 1 to 5



1

just learning about KEEN



5

I could lead this workshop

Group Activity

- Each group will receive customer support phone call information in the form of features, along with definitions for each feature.



[This Photo](#) by Unknown Author is licensed under [CC BY-NC](#)

Example Information

Features (Label hidden)

Feature	Value
Call Failure	14.0
Complaints	0.0
Subscription Length	40.0
Charge Amount	3.0
Seconds of Use	7515.0
Frequency of use	103.0
Frequency of SMS	201.0
Distinct Called Numbers	28.0
Age Group	3.0
Tariff Plan	1.0
Status	1.0
Age	30.0
Customer Value	1108.72

Example Information

Data Dictionary

Column	Explanation
Call Failure	number of call failures
Complaints	binary (0: No complaint, 1: complaint)
Subscription Length	total months of subscription
Charge Amount	ordinal attribute (0: lowest amount, 9: highest amount)
Seconds of Use	total seconds of calls
Frequency of use	total number of calls
Frequency of SMS	total number of text messages
Distinct Called Numbers	total number of distinct phone calls
Age Group	ordinal attribute (1: younger age, 5: older age)
Tariff Plan	binary (1: Pay as you go, 2: contractual)
Status	binary (1: active, 2: non-active)
Age	age of customer
Customer Value	the calculated value of customer

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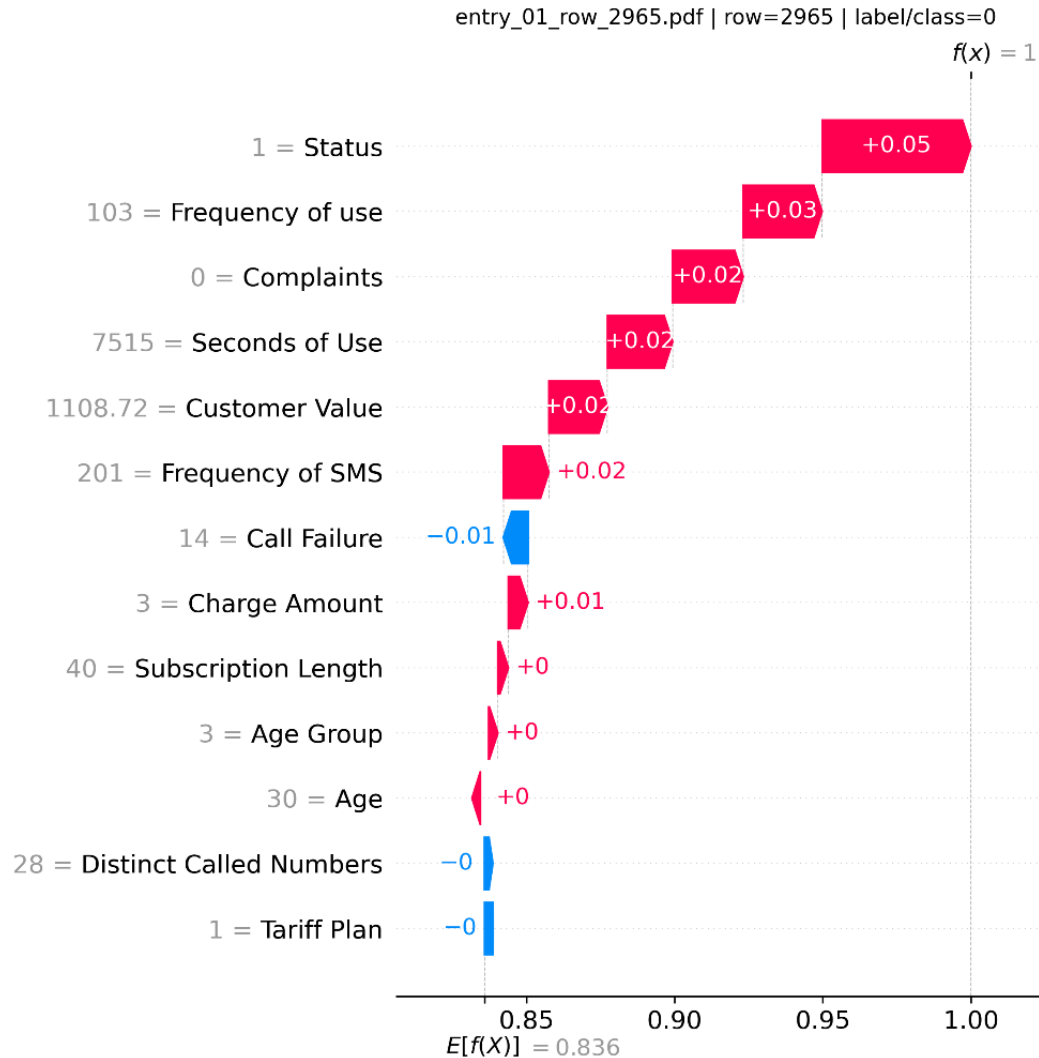
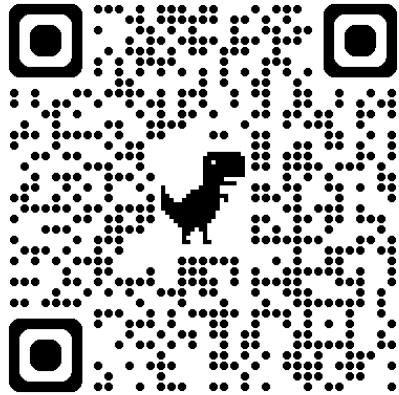
Group Activity

- Look at your specific customer data and discuss with your group whether this customer will churn or not.
- One person should aggregate the discussed information and report the group's final decision to everyone.

Group Activity

- Each group will now receive information about feature impact on the decision.
- Reflect as a group on your initial assumptions and analyze how each feature influences the final outcome.

Example Information



<https://colab.research.google.com/drive/11WQdWFnpfsMjuws-hFc8ZyZ0pKPFjO0k>

How could **XAI** tools be used to promote Connections?

Choose a technical learning outcome (LO) from your course



Pick an assignment related to this LO from your course/project

Which EM learning outcomes best align with your revised course assignment?

Connections

- a. Understand ramifications (technical and non-technical) of design decisions.
- b. Identify and evaluate sources of information.
- c. Connect life experiences with class content.
- d. Connect content from multiple courses to solve a problem.
- e. Integrates/synthesizes different kinds of knowledge.
- f. Consider a problem from multiple viewpoints.
- g. Promotes responsibility from multiple perspectives (societal, financial, environ., etc...)
- h. Articulates the idea to diverse audiences.
- i. Understands how elements of an ecosystem are connected.
- j. Identifies and works with individuals with complementary skill sets, expertise, etc.
- k. Develop a professional network.
- l. Demonstrate the ability to consider risk as a factor.



Co Creation with AI



Creating Value

Seek opportunities. Understand stakeholders. Have a positive impact.

Each group select one of the following entrepreneurial tasks

- Identify a problem worth solving for a defined group of people
- Propose a new product, service, or experience
- Redesign an existing process to improve efficiency, quality, or access



Each group must identify a stakeholder

- Users / beneficiaries
- Students / learners
- Faculty / staff
- Organization / institution
- Industry partners
- Community / society
- Or use your Own!





<https://gemini.google.com/gem/33a41862bd77>

Why, what and How. Explain it please!

- Why this solution was chosen over alternatives?
- What assumptions you are making about each stakeholder
- How value is created for each stakeholder?
- Where uncertainty, risk, or trade-offs exist?
- What parts require human judgment or oversight?



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How!



Start with a human-written prompt naming the challenge and stakeholders



Ask AI to generate or refine solution concepts



Make at least one human modification based on stakeholder value or trade-offs

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How could **AI Brainstorming** be used to Create Value?

Choose an engineering design learning outcome (LO) from
your course



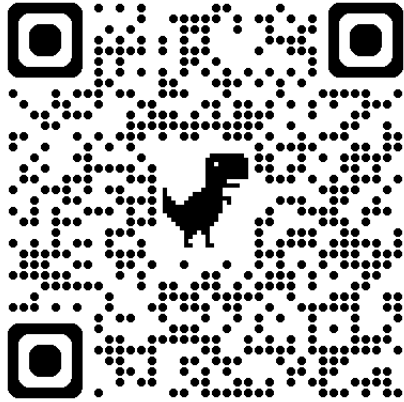
Pick an assignment related to this LO from your course/project

Which EM learning outcomes best align with your revised course assignment?

Creating Value

- a. Identify the needs and motivations of various stakeholders.
- b. Express empathy in identifying problems and exploring solutions.
- c. Create solutions that meet customer needs.
- d. Defines a market and market opportunities
- e. Craft a compelling value proposition tailored to specific stakeholders.
- f. Integrate non-monetary and monetary factors into a triple bottom line assessment.
- g. Applies technical skills/knowledge to the development of a technology/product
- h. Modifies an idea/product based on feedback
- i. Focuses on understanding the value proposition of a discovery
- j. Describes how a discovery could be scaled or sustained, using elements such as revenue streams, key partners, costs, and key resources.
- k. Engages in actions with the understanding that they have the potential to lead to both gains or losses.

Workshop and all handouts are in our card!



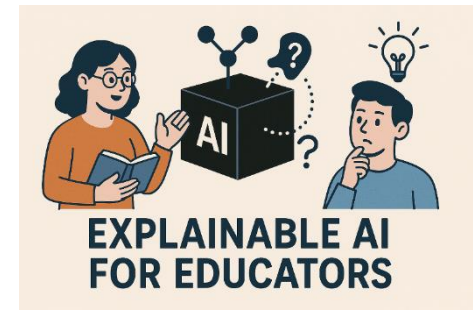
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**For more information please
reach out to**

**Mohammad El-Yabroudi and
Eric G. Meyer**

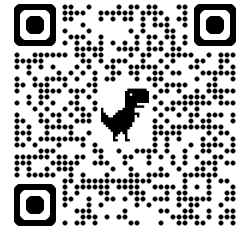
melyabrou@ltu.edu
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Links



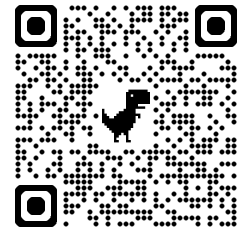
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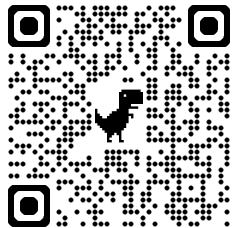


<https://playground.tensorflow.org/>

<https://notebooklm.google.com/notebook/29cec7e7-5246-4a51-93c9-5bf396097a3f>



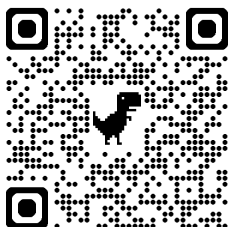
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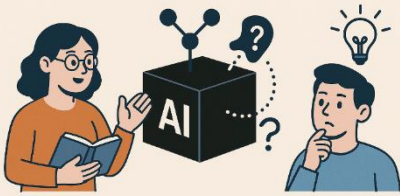
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EXPLAINABLE AI FOR EDUCATORS

Engaging Students and Inspiring Entrepreneurial Mindsets

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XAI & EM Learning Module Development Worksheet

Notes:

Module 1: GenAI & Student Reflection (Curiosity)

Step 1: Choose the Technical LO

Choose an Existing Assignment

Step 2: How could GenAI tools be used to increase reflection?

Step 3: Alignment
with an EM LO:

Reflection & Next Steps:

Which intervention seems the most “classroom ready”?

Module 2: XAI & Connections

Step 1: Choose the Technical LO

Choose an Existing Assignment

Step 2: How could XAI tools improve connections?

Step 3: Alignment
with an EM LO:

Module 3: AI Brainstorming & Creating Value

Step 1: Choose the Engineering Design LO

Choose an Existing Assignment

Step 2: How could GenAI tools be used to create value for stakeholders?

Step 3: Alignment
with an EM LO:

2026

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Synthesis and Consensus

"Based on the paper and our discussion, what were the three biggest unresolved questions identified by the group? What follow-up study did we agree was necessary?"

- Creates an archive of the intellectual journey, allowing students to track their critical appraisal skills over the semester.

"Discuss how datasets (e.g., Caucasian male knees) might fail to accurately diagnose or design for other demographics."

- Uses AI to analyze diverse datasets to reveal these biases.

Assessment

To effectively use AI as a learning tool, assignments must move beyond simple output generation. The focus must be on the process of inquiry and the evaluation of outputs.

Verification & Accuracy	Identify AI hallucinations; explicitly verify AI claims against primary text with precise citations.
Prompt Engineering	Use sophisticated, iterative prompting (Chain-of-Thought) to drill down into methodology/controls.
Critical Synthesis	Synthesize AI output with deep human insight; identify gaps the AI missed; connect to broader clinical context.
Ethical Reflection	Nuanced reflection on AI limitations (bias, safety) in clinical decision-making.